

EXAMPLE 12.4

Determine the reaction forces and the driving torque for the four-bar linkage in the posture shown in Fig. 12.9. The input crank is operating at a constant angular velocity $\omega_2 = 48 \text{ rad/s}$ ccw, and the applied force acting at point C is $\mathbf{F}_C = -0.8\hat{j} \text{ kN}$. The masses of links 3 and 4 are $m_3 = 1.5 \text{ kg}$ and $m_4 = 5 \text{ kg}$, respectively, and the second moments of mass of links 2, 3, and 4 are $I_{G_2} = 0.025 \text{ kg} \cdot \text{m}^2$, $I_{G_3} = 0.012 \text{ kg} \cdot \text{m}^2$, and $I_{G_4} = 0.054 \text{ kg} \cdot \text{m}^2$, respectively.

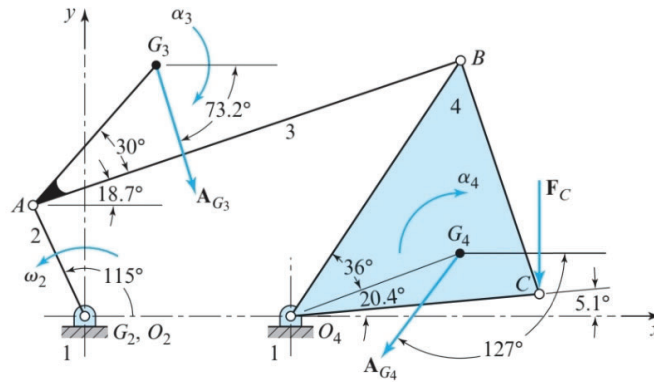


Figure 12.9 $R_{AO_2} = 60 \text{ mm}$, $R_{O_4O_2} = 100 \text{ mm}$, $R_{BA} = 220 \text{ mm}$, $R_{BO_4} = 150 \text{ mm}$, $R_{CO_4} = R_{CB} = 120 \text{ mm}$, $R_{G_3A} = 90 \text{ mm}$, and $R_{G_4O_4} = 90 \text{ mm}$.

Assumptions: Friction and gravity can be neglected.

Also Given, The results of Kinematic Analysis $\rightarrow \vec{\omega}_3 = 18.2719 \hat{k} \text{ rad/s}$, $\vec{\omega}_4 = 31.2072 \hat{k} \text{ rad/s}$
 $\vec{\alpha}_3 = -119 \hat{k} \text{ rad/s}^2$, $\vec{\alpha}_4 = -625 \hat{k} \text{ rad/s}^2$

Step 1: Compute acceleration of COM of each link and the angular acceleration of each link.

$$\vec{a}_{G_3} = \vec{a}_A - \omega_3^2 \vec{r}_{G_3/A} + \vec{\alpha}_3 \times \vec{r}_{G_3/A}$$

$$\vec{a}_A = -\omega_2^2 \vec{r}_{A/O_2}$$

$$\Rightarrow \vec{r}_{G_3/A} = (0.090 \cos 48.7^\circ \hat{i} + 0.090 \sin 48.7^\circ \hat{j}) \text{ m} = (0.0594 \hat{i} + 0.0676 \hat{j}) \text{ m}$$

$$\Rightarrow \omega_2 = 48 \hat{k} \text{ rad/s}, \quad \vec{r}_{A/O_2} = (0.060 \cos 115^\circ \hat{i} + 0.060 \sin 115^\circ \hat{j}) = (-0.0254 \hat{i} + 0.0544 \hat{j}) \text{ m}$$

$$\alpha_3 = -119 \hat{k} \text{ rad/s}^2$$

$$\Rightarrow \vec{a}_A = (58.42 \hat{i} - 125.29 \hat{j}) \text{ m/s}^2$$

$$-\omega_3^2 \vec{r}_{G_3/A} = -(18.2719)^2 (0.0594 \hat{i} + 0.0676 \hat{j}) \text{ m/s}^2$$

$$\vec{\alpha}_3 \times \vec{r}_{G_3/A} = -119 \hat{k} \times (0.0594 \hat{i} + 0.0676 \hat{j}) \text{ m/s}^2$$

$$\boxed{\vec{a}_{G_3} = (46.64 \hat{i} - 154.93 \hat{j}) \text{ m/s}^2}, \boxed{\vec{\alpha}_3 = -119 \hat{k} \text{ rad/s}^2}$$

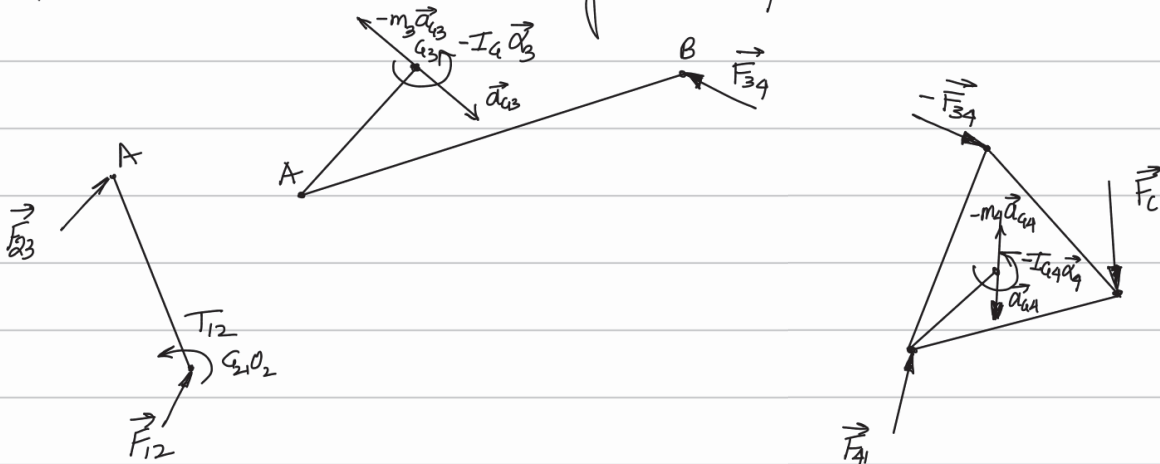
$$\vec{a}_{G_4} = -\omega_4^2 \vec{r}_{G_4/O_4} + \vec{\alpha}_4 \times \vec{r}_{G_4/O_4}$$

$$\vec{r}_{G_4/O_4} = (0.0844 \hat{i} + 0.0314 \hat{j}) \text{ m}$$

$$\Rightarrow \vec{a}_{G_4} = -31.2072^2 (0.0844 \hat{i} + 0.0314 \hat{j}) + (-625 \hat{k}) \times (0.0844 \hat{i} + 0.0314 \hat{j})$$

$$\boxed{\vec{a}_{G_4} = (-62.55 \hat{i} - 83.27 \hat{j}) \text{ m/s}^2} \quad \& \quad \boxed{\alpha_4 = -625 \hat{k}}$$

Step 2: Draw the F.B.D.'s including inertia force and inertia moments.



Step 3: Write equilibrium equations for each link

For link 2: $\sum \vec{F} = 0, \quad \sum \vec{M}_{O_2} = 0$

$$\vec{F}_{12} + \vec{F}_{23} = 0 \rightarrow (1)$$

$$\sum \vec{M}_{O_2} = \vec{r}_{A/O_2} \times \vec{F}_{23} + T_{12} = 0 \rightarrow (2)$$

For link 3: $\sum \vec{F} = 0$ and $\sum \vec{M}_A = 0$

$$-\vec{F}_{23} + \vec{F}_{34} + (-m \vec{a}_{G3}) = 0 \rightarrow (3)$$

$$\Rightarrow \sum \vec{M}_A = \vec{r}_{G3/A} \times (-m \vec{a}_{G3}) + \vec{r}_{B/A} \times \vec{F}_{34} + (-I_{G3} \vec{\alpha}_3) = 0 \rightarrow (4)$$

For link 4: $\sum \vec{F} = 0$ and $\sum \vec{M}_Q = 0$

$$\Rightarrow -\vec{F}_{34} + \vec{F}_C + \vec{F}_A + (-m_4 \vec{a}_{G4}) = 0 \rightarrow (5)$$

$$\Rightarrow \sum \vec{M}_Q = \vec{r}_{B/Q} \times (-\vec{F}_{34}) + \vec{r}_{G4/Q} \times (-m_4 \vec{a}_{G4}) + \vec{r}_{C/Q} \times \vec{F}_C + (-I_{G4} \vec{\alpha}_4) = 0 \rightarrow (6)$$

9 equations: (1)x, (1)y, (2), (3)x, (3)y, (4), (5)x, (5)y, (6)

9 unknowns: $F_{12x}, F_{12y}, F_{23x}, F_{23y}, F_{34x}, F_{34y}, F_{A1x}, F_{A1y}, T_{12}$

$\vec{F}_{34}$ can be determined from eqn. (4) and eqn. (6)

Once $\vec{F}_{34}$ is known

Compute $\vec{F}_{23}$ from eqn. (3)

Compute T_{12} from eqn. (2)

Compute $\vec{F}_{12}$ from eqn. (1)

Compute $\vec{F}_A$ from eqn. (5)

eqn. (4)

$$\sum \vec{M}_A = \vec{r}_{G3/A} \times (-m \vec{a}_{G3}) + \vec{r}_{B/A} \times \vec{F}_{34} + (-I_{G3} \vec{\alpha}_3) = 0 \rightarrow (4)$$

$$\Rightarrow \vec{r}_{B/A} = 0.220 (\cos 18.7^\circ \hat{i} + \sin 18.7^\circ \hat{j}) = (0.2084 \hat{i} + 0.0705 \hat{j}) \text{ m}$$

$$\Rightarrow (0.0594 \hat{i} + 0.0676 \hat{j}) \times (-1.5 (46.64 \hat{i} - 154.93 \hat{j})) \\ + (0.2084 \hat{i} + 0.0705 \hat{j}) \times (F_{34x} \hat{i} + F_{34y} \hat{j}) + (-0.012 (-119) \hat{k}) = 0$$

$$\Rightarrow \boxed{0.07053 F_{34x} - 0.20839 F_{34y} = -26.035} \rightarrow (7)$$

From equation (6)

$$\Rightarrow \vec{r}_{B/O_4} \times (-\vec{F}_{34}) + \vec{r}_{G_4/O_4} \times (-m_4 \vec{a}_{G_4}) + \vec{r}_{C/O_4} \times \vec{F}_C + (-I_{G_4} \vec{\alpha}_4) = 0$$

$$\Rightarrow \vec{r}_{B/O_4} = (0.15 \cos 56.4^\circ \hat{i} + 0.15 \sin 56.4^\circ \hat{j}) = (0.083 \hat{i} + 0.12494 \hat{j}) \text{ m}$$

$$\Rightarrow (0.083 \hat{i} + 0.12494 \hat{j}) \times (-F_{34x} \hat{i} - F_{34y} \hat{j}) + (0.084355 \hat{i} + 0.03137 \hat{j}) \times (-5(-62.54 \hat{i} - 83.27 \hat{j})) \\ + (0.11952 \hat{i} + 0.010667 \hat{j}) \times (-800 \hat{j}) - (0.054) (-625 \hat{k}) = 0$$

$$\Rightarrow \boxed{-0.12494 F_{34x} + 0.083 F_{34y} = -36.5542} \rightarrow (8)$$

Solve the system of linear equations (7) & (8) to get

$$F_{34x} = 295.39 \text{ N} \quad \& \quad F_{34y} = 4.19 \text{ N} \Leftrightarrow \vec{F}_{34} = (295.39 \hat{i} + 4.19 \hat{j}) \text{ N}$$

$$\begin{aligned} 0.07053 F_{34x} - 0.20839 F_{34y} &= -26.033 && \rightarrow \text{from eqn. 4} \\ -0.12494 F_{34x} + 0.083 F_{34y} &= -36.5542 && \rightarrow \text{from eqn. 6} \end{aligned}$$

$$F_{34x} = 295.34 \quad \& \quad F_{34y} = 4.1740$$

$$\boxed{\vec{F}_{34} = 295.34 \hat{i} + 4.1740 \hat{j}} \text{ N}$$

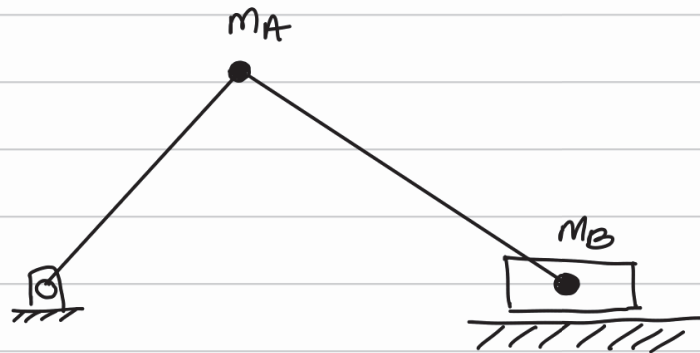
$$\vec{F}_{23} = 225.4309 \hat{i} + 236.5836 \hat{j} \text{ N}$$

$$\Rightarrow \vec{T}_{12} = -225.4309 \hat{i} - 236.5836 \hat{j} \text{ N}$$

$$\Rightarrow \vec{T}_{41} = -17.3414 \hat{i} - 387.8158 \hat{j} \text{ N}$$

$$\Rightarrow \vec{T}_{12} = 18.2577 \hat{k} \text{ Nm}$$

Shaking forces and moments



$$\Rightarrow \vec{f}_s = \vec{f}_A + \vec{f}_B = R\omega^2 \left[(m_A + m_B - m_{cb}) \cos\theta + m_B \frac{R}{L} \cos 2\theta \right] \hat{i} \\ + R\omega^2 (m_A - m_{cb}) \sin\theta \hat{j}$$

1. If we have n cylinders, the inertia force associated with each piston is in the direction of piston travel and is given by

$$f_{Bi} = m_B R \omega^2 \left[\cos(\theta + \phi_i) + \frac{R}{L} \cos 2(\theta + \phi_i) \right], \quad i=1, 2, \dots, n$$

$\phi_i \rightarrow$ phase angle of the i^{th} reciprocating mass.

$\theta \rightarrow$ crank angle (for the first cylinder).

This formula is valid for in-line engines.